

IT3051 - Fundamentals of Data Mining
Mini Project 2026 - Dataset Proposal



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1. Problem Scenario

An online retailer wants to know if the current session is likely to lead to a purchase while a customer is still browsing the site. Such prediction can be used by Marketing, UX and CRM teams to trigger timely interventions (e.g. a discount prompt, live-chat offer, exit-intent email capture) for sessions showing signs of abandonment and to better understand which browsing behaviors drive conversion.

Prediction objective	Predict whether a browsing session will end in a purchase (session-level purchase intent).
Task type	Binary classification.
Stakeholders	E-commerce marketing team, UX/product design team, CRM and retention team, site analytics team.
Decisions supported	Real-time on-site interventions (offers, pop-ups, chat prompts), personalization of content, prioritization of UX improvements on pages most linked to abandonment.
System inputs	Session-level browsing behavior available before the outcome is known: page-visit counts and durations by category, bounce/exit rates, page value, proximity to a special day, month, visitor type, weekend flag, and technical/traffic metadata (operating system, browser, region, traffic source).
System output	Predicted purchase probability and a Yes/No classification for the session, optionally with the top contributing factors.

2. Dataset Identification

Dataset name	Online Shoppers Purchasing Intention Dataset
Original source	UCI Machine Learning Repository, contributed by C. Okan Sakar (Bahçeşehir University) and Yomi Kastro (Inveon Information Technologies).
Dataset URL	https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset
Citation	Sakar, C. & Kastro, Y. (2018). Online Shoppers Purchasing Intention Dataset [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C5F88Q .
Associated research paper	Sakar, C.O., Polat, S.O., Katircioglu, M. & Kastro, Y. (2019). Real-time prediction of online shoppers' purchasing intention using multilayer perceptron and LSTM recurrent neural networks. Neural Computing and Applications, 31, 6893–6908.
License	Creative Commons Attribution 4.0 International (CC BY 4.0) — permits sharing and adaptation with attribution.

3. Dataset Description and Real-World Context

The dataset includes feature vectors of 12330 online shopping sessions over a year from a real e-commerce website. Each row represents a unique session from a different visitor, deliberately sampled to avoid any bias toward a single campaign, special day, user profile or time period. The features encode a visitor's behavior on the site during a session, including the number of pages of each type viewed, the duration of the session, and common web-analytics measures (bounce rate, exit rate, page value) collected by Google Analytics for the pages visited, as well as contextual factors like the month, closeness to a special day (say, Valentine's Day), visitor type, and whether the session took place on a weekend. This directly matches the assigned scenario: predicting purchase intent from real, observable browsing behavior on an e-commerce platform.

4. Dataset Characteristics

Number of records	12,330 sessions
Number of features	17 predictor attributes + 1 target variable (18 columns total): 10 numerical and 8 categorical/boolean attributes.
Relevant features	Administrative, Administrative_Duration, Informational, Informational_Duration, ProductRelated, ProductRelated_Duration (page-visit counts and time spent per page category); BounceRates, ExitRates, PageValues (Google Analytics session-quality metrics); SpecialDay (closeness to a special day); Month, Weekend, VisitorType (contextual attributes); OperatingSystems, Browser, Region, TrafficType (technical/traffic metadata).

5. Target Variable

Target variable	Revenue
Data type	Boolean (True / False)
Meaning	Indicates whether the browsing session ended with the visitor completing a purchase (True) or not (False).
Class distribution	10,422 negative sessions – no purchase ($\approx 84.5\%$) vs. 1,908 positive sessions – purchase made ($\approx 15.5\%$). This is an imbalanced binary target.

6. Expected Machine Learning Task

Binary classification – predicting the two classes of Revenue (purchase vs no purchase) from behavioral and contextual features of the session. Because of the class imbalance, the evaluation will be based on precision, recall, F1-score and ROC/PR-AUC, rather than accuracy alone. During the preprocessing and modeling, resampling or class-weighting strategies will be evaluated.

6. Reason Why This Dataset Is Suitable

- Directly relevant: it is a genuine session-level e-commerce dataset whose behavioral features match exactly what the assigned scenario needs (real-time purchase-intent prediction).
- Clear target variable: Revenue is unambiguous and directly usable for supervised classification.
- Sufficient scale and richness: 12,330 records and 17 features (mixed numerical and categorical) support meaningful EDA, feature engineering, multiple modelling approaches, and hyperparameter tuning.
- Credible, citable source: published via the UCI Machine Learning Repository and an associated peer-reviewed paper, allowing proper academic citation.
- Realistic complexity: natural class imbalance, skewed numerical distributions, and mixed anonymized categorical codes give genuine opportunities to justify preprocessing decisions rather than working with an already clean toy dataset.
- Accessible to the whole group: freely downloadable (CSV/XLSX) with no registration, licensing fee, or restricted-access barrier.

8. Data-Quality Observations and Known Limitations

- No missing values are present in any of the 18 columns, so no imputation is required.
- Strong class imbalance (~84.5% vs ~15.5%), which must be addressed through resampling (e.g. SMOTE) and/or class-weighted models, and through imbalance-aware evaluation metrics.
- Several numerical features (e.g. Administrative_Duration, ProductRelated_Duration, BounceRates) are heavily right-skewed with many zero values, which may require transformation, outlier capping, or robust scaling.
- BounceRates and ExitRates are conceptually and statistically correlated, which will be examined during EDA and may warrant dropping or combining one of them to reduce redundancy.
- OperatingSystems, Browser, Region, and TrafficType are provided as anonymized integer codes rather than descriptive labels, so they must be treated as nominal categories (not ordinal) during encoding.
- The data covers a single e-commerce site over a single year, so the model's generalizability to other platforms, markets, or time periods is limited and will be noted as a limitation in the final report.

9. Ethical, Privacy, Licensing, and Accessibility Considerations

- Privacy: the dataset contains no personally identifiable information (no names, emails, IP addresses, or account IDs). All fields are session level behavioral or technical metrics, so privacy risk is minimal.
- Licensing: released under a CC BY 4.0 license, which permits academic use, adaptation, and redistribution provided the original authors are credited (citation included in Section 2).
- Accessibility: publicly downloadable in CSV format from the UCI Machine Learning Repository with no cost, registration, or access restriction, so it is equally accessible to every group member.
- Ethical use: intended use (supporting marketing/UX decisions) is a standard, non manipulative e-commerce analytics application; the project will avoid any framing that encourages dark pattern or deceptive interventions based on the model's predictions.

10. Summary

The Online Shoppers Purchasing Intention Dataset is a trustworthy and publicly available dataset that is complex enough to support the assigned real-time purchase-intent prediction scenario. It gives a clean binary target variable, a realistic mix of numerical and categorical features, real data-quality issues (imbalance, skew, correlated features) that require meaningful preprocessing decisions, and no ethical or privacy constraints on its use for academic purposes.